Metallurgy of copper:

Copper:-

Copper is a redish element, extremly ductile metal of group 11 of the periodic table that is unusually good conductor of heat and electricity.

Ores for metallurgy of copper:

The cheif ores for metallurgy of copper are as follows:

- Copper pyrites $CuFeS_2$
- Cuprite or ruby copper Cu_2O
- Copper glance Cu_2S
- Malachite $Cu(OH)_2 . CuCO_3$
- Azurite $2CuCO_3 \cdot Cu(OH)_2$.

Presence:

Copper is found in native U.S.A. Mexico, Russia and China.

Extraction of copper:

The copper can be extracted by 2 processes:-

- Dry process
- Wet process

The dry process includes the extraction from oxide and carbonate ores, from copper pyrites. It includes Roasting, Smelting and Bessemerisation.

The wet process includes the sulphate roasting, hydrometallurgical process, sulphuric acid leaching.

• Dry process

(i) From oxides and carbonate ores:-

Firstly, the ore is crushed, concentrated by gravity process and roasted. The carbonate decompose and form oxides

$$CuCO_3 . Cu(OH)_2 \longrightarrow 2\,CuO + CO_2 + H_2O.$$

Malachite

The oxide that is formed reduce with carbon in reverberatory furnace

$$CuO + C \rightarrow Cu + CO$$

(ii) From copper pyrites:-

This is the cheif ore of copper and from that, the extraction involves following steps:

* Concentration
* Roasting
* Smelting
* Bessmerization.

1- Concentration:

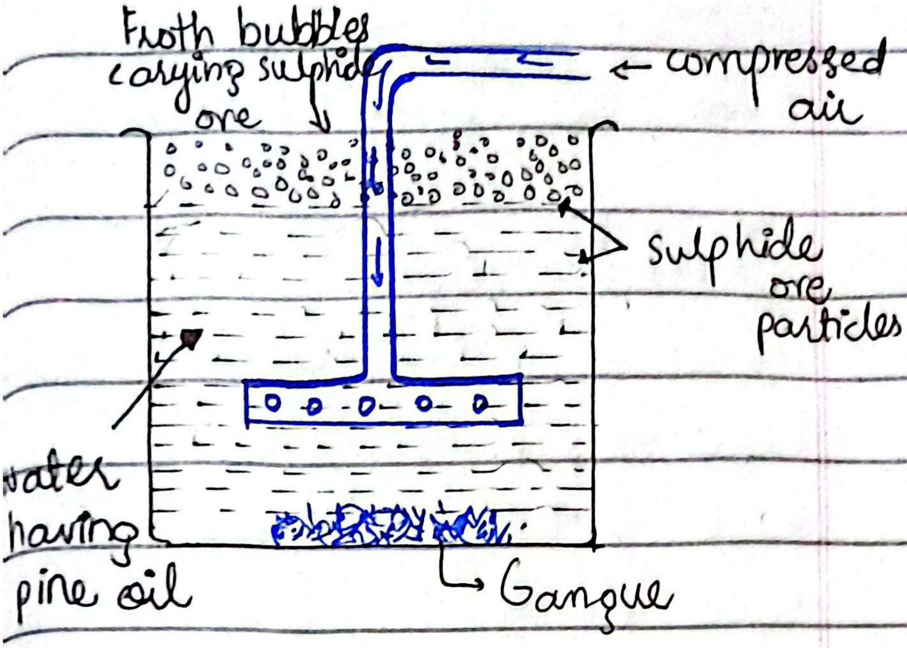
Copper pyrite, naturally associated with the large amount of silicious impurities. To remove all these impurites, the crushed copper first crushed, seived and then concentrated by the froth flotation process.

Froth flotation process:

Froth flotation process is based on the wetting characteristic of ore and gangue particles with oil and water respectively.

Process:

The ore particles are wetted by pine oil and gangue particles by water. The whole mixture is agitated by compressed air. Hence, oil coated particles become lighter and come on the surface of (water) mixture in the form of froth that can be skimmed.



Roasting:-

It is the process of heating the concentrated ore to a high temprature in the excess of air.

e.g.

Copper pyrite is strongly heated in excess of air to convert it into the mixture of cuprous sulphide and ferrous sulphide. $[Cu_2S + FeS]$, while impurities react with oxygen to form oxides.

Rex.

$$2CuFeS_2 + O_2 \rightarrow Cu_2S + 2FeS + SO_2$$

Smelting:-

It is the further heating of the roasted ore with the sand flux and coke in the presence of excess of air in the Blast furnace. The blast furnace is about **15-20 ft** in height and is made up of sheet steal lined with the refractory bricks. The process is highly exo-thermic, therefore a small amount of coke is required in this process. In this process, first ferrous sulphide is oxidized to form ferrous oxide, which react with sand to form iron silicate ~~sand~~

slag. It being lighter to the top and is removed from the upper hole.

Rex.

$$2FeS + 3O_2 \rightarrow 2FeO + 2SO_2 \uparrow$$

$$FeO + SiO_2 \rightarrow FeSiO_3.$$

On the other hand, cuprous sulphide also oxidizes to form cuprous oxide which react with unreacted ferrous sulphide to form ferrous oxide and cuprous sulphide. In this way cuprous sulphide and ferrous sulphide form a mixture ($Cu_2S \cdot FeS$)

Matte:-

The mass so produced contains mostly cuprous sulphide with a little FeS. It is called matte. It is withdrawn from the lower hole. It contains about **45%** of copper.

$$2Cu_2S + 3O_2 \longrightarrow 2Cu_2O + 2SO_2$$
$$Cu_2O + FeS \longrightarrow Cu_2S + FeO$$

Bassmerization:-

It is the further heating of molten matte in a pear shaped bessmer convertor. It is fixed on a pivot, so it can be tilted in any direction. Molten matte is mixed with sand and heated with a hot blast of air through tueyers. Ferrous sulphide is oxidized to form ferrous oxide. Which reacts with sand to form slag ($FeSiO_3$) that floats on the top.

$$2FeS + 3O_2 \rightarrow 2FeO + 2SO_2$$
$$FeO + SiO_2 \longrightarrow FeSiO_3$$

On the other hand, cuprous sulphide is oxidized to form cuprous oxide, which again react with remaining cuprous sulphide to form metallic copper.

$$2Cu_2S + 3O_2 \longrightarrow 2Cu_2O + 2SO_2\uparrow$$

$$2Cu_2O + Cu_2S \longrightarrow 6Cu + SO_2\uparrow$$

Blister copper:-

The molten matte is shifted from the convertor to sand moulds and is allowed to cool. The dissolved gases escape out forming blisters on the surface of solid copper. Therefore, it is called blister copper. It is about 98% pure copper. It is further refined by electrolysis.

Refining and purification of metal:-

Refining of impure metal by electrolysis is the most widely used process.

1- Electrolytic refining:
example:-

Electrolytic refining of copper is carried out in an electrolytic tank having copper sulphate solution. Two electrodes, one impure copper metal that acts as a anode and other pure copper metal act as cathode are suspended in the electrolytic solution.

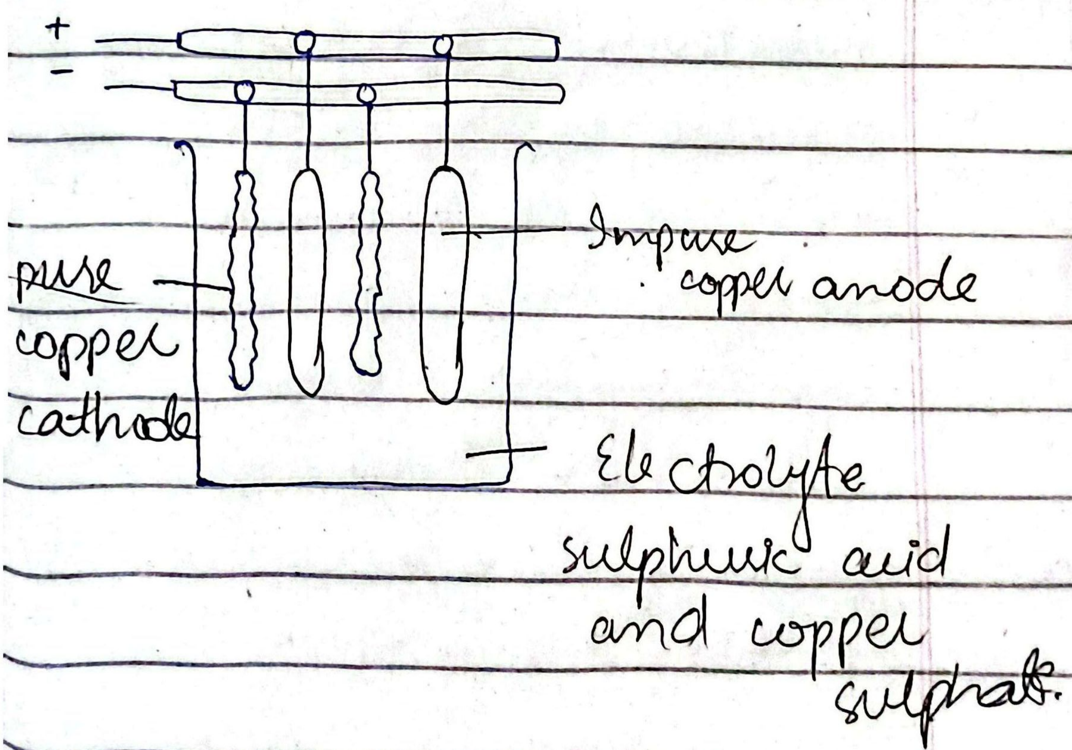
On passing electric current through solution, anode dissolves to provide Cu^{2+} ions to solution. These Cu^{2+} ions are discharged by gaining of electrons from the cathode. Thereby, copper atoms deposit on the cathode, making it thick block of pure copper metal.

Anode mud:-

The impurities

like gold and silver act as anode mud.

In the process, impure copper from the anode dissolves and goes into the copper sulphate solution. Side by side, pure copper ions from the solution deposit on the cathode. Thus, the cathode become pure copper metal. The impurities like gold and silver settle down as anode mud.



2- Poling:

The blister copper is melted on the hearth of reverberatory furnace. The oxygen present in the metal as Cu_2O oxidizes impurities. For example, iron is oxidized to FeO which forms a slag with silica lining of furnace. The oxides of others, pass out as gas or form a scum that can be removed. Excess of Cu_2O renders copper brittle. This Cu_2O excess is removed by covering the surface of fused metal with powdered anthracite and stir with poles of green wood. Then the copper is 99.5% pure.

Wet process:-

In this process, metal is first converted into some of it's soluble salts generally sulphates or chlorides. From the solution so obtained, copper is obtained by precipitation with iron or by electrolysis

Method's :-

The methods for wet process are as follows:-

- Sulphate roasting
- Hydrometallurgical process
- Sulphuric acid leaching

Sulphate roasting:-

The ore is calcined in a reverberatory furnace as a result of which the sulphide of copper is converted into copper sulphate.

$$2Cu_2S + 5O_3 \rightarrow 2CuSO_4 + 2CuO$$

$$CuO + SO_3 \rightarrow CuSO_4.$$

- SO_3 is produced by the decomposition of $FeSO_4$ formed by the calcination of FeS.
- $CuSO_4$ is extracted with water and copper recovered from it by the addition of scrap iron.

$$CuSO_4 + Fe \rightarrow Cu + FeSO_4$$

→ Hydrometallurgical process:-

Heaps ~~and~~ of pyrites are exposed to air and rain. The sulphide ore is oxidized to copper sulphate, iron sulphate and sulphuric acid.

→ Sulphuric acid leaching:-

The native copper, non-sulphide ores and sometimes sulphide ores are leached with dilute sulphuric acid when a solution of copper sulphate is obtained.

Physical properties of copper

The physical properties of copper are as follows:-

Electronegativity and Atomic radius:-

→ Its electronegativity is 1.9 and atomic radius is 128 pm.

→ The first three ionization energies values of Cu are 745.3, 1957.3 and 3577.6 kJ mol^{-1}

→ Copper is reddish brown metal which melts at 1356 K.

→ It is malleable and ductile.

It is an excellent conductor of heat and electricity, the best or next conductor to silver.

The values of heat of fusion and vapourization are 13.0 and 307 kJ mol^{-1}.

Uses of copper:-

- It is used for making electric cables and other electrical applinces.
- Copper sheets are used for making utensils.
- It is used for the prep-ration of various salts which are used as insecticides, pesticides, pigments etc.
- It is used in electro-plating, electrotyping and in making coins.
- Copper forms a number of alloys like brass, bronze etc. Use in arts and industry.